0580_w24_ms_41_Q1

Question	Answer	Marks	Partial Marks	Guidance
1(a)(i)	2 × 5 × 7 [=70]	**2**	**B1** for 2, 5, 7	B1 may be seen on a diagram. Eg factor tree Allow B1 for inclusion of 1 with correct factors and no other extras
1(a)(ii)	14	**2**	**M1** for [112 =] $2^4 \times 7$ oe or for answer 2 × 7	M1 allow prime factors not as a product eg in a factor tree, repeated division For M1 condone inclusion of 1 and no extras
1(a)(iii)	$560x^4y^5$	**2**	**B1** for answer kx^4y^5 or for answer $560x^ay^b$ or for correct answer seen then spoiled	For B1 accept $k = 1$ or k a product of integers and accept answer 560. Condone × signs for 2 marks or B1
1(b)(i)	a^8	**1**		
1(b)(ii)	$\frac{c}{8}$ final answer	**2**	**M1** for $\frac{5bc}{40b}$ or better	For 2 marks accept answer $\frac{1}{8}c$, $\frac{1c}{8}$, but 1/8c only if seen correct in working M1 is for a single fraction. After M1 awarded isw incorrect cancelling. ½ x c/4 not enough for M1. Condone × signs for M1 but not 2 marks.
1(c)	5.5 or $\frac{11}{2}$ or 5½	**2**	**M1** for $2x = 15 - 4$ oe or $2 + x = \frac{15}{2}$ oe	M1 is for isolating the x term correctly or making coefficient of $x = 1$. Allow $-2x = 4 - 15$ oe
1(d)	–2	**3**	**M1** for $34 + 2x = 5(4 - x)$ oe or better **M1** dep for reaching $ax = b$ FT *their* first step	M1 for clearing fraction. $6.8 + 0.4x = 4 - x$ M1 for isolating x term. FT dep on 4 term expression eg. $34 + 2x = cx + d$ or $ex + f = 4 - x$. Allow recovery to $34 + 2x = 20 - 5x$ after omission of brackets.

1(e)(i)	11	**2**	**M1** for $7 + \sqrt[3]{(-8)^2}$ oe	M0 for $7 + \sqrt[3]{-8^2}$ but allow recovery. Do not allow misread of cube root as square root.
1(e)(ii)	$[\pm]\sqrt{(P-d)^3}$ oe final answer	**3**	**B1** for $P - d = \sqrt[3]{m^2}$ oe **M1** for cube both sides **M1** for square root leading to final answer	Do not allow misread of cube root as square root For 3 marks $[\pm]\ (P-d)^{3/2}, -(P-d)^{3/2}$ $P - d = m^{2/3}$ oe is for isolating m term 2nd and 3rd methods can be done in either order, FT from their first step. Award a maximum of 1 out of 3 marks if no attempt has been made to both cube and square root (unless B1 M1 has already been awarded for two correct steps) Allow notation $\sqrt[2]{x}$ for $\sqrt{x}$ **See AG**